REPORT

W&B dashboard: forge.coreweave.com/wandb/gkobea-hamilton-college/oxpet-unetpp

Summary

Strong data augmentation was the change that mattered. It raised test mIoU from 0.816 to 0.829 and stopped the baseline from overfitting, with most of the gain on the hard border class (IoU 0.615 to 0.645). Adding a Dice term to the cross-entropy loss gave no measurable further gain (0.829 to 0.829). Horizontal-flip test-time augmentation (TTA) added 0.4 to 0.6 mIoU points to every model at no training cost. The best model (strong augmentation, CE+Dice, flip TTA) reaches 0.833 mIoU and 0.903 mean Dice on the official test split (3,669 images).

Setup

Task and data. Three-class trimap segmentation. Trimap values 1/2/3 are mapped to classes 0 = pet, 1 = background, 2 = border. The official trainval list (3,680 images) is split 80/20 with a fixed seed into 2,944 training and 736 validation images. The official test split is used only for final reporting. We checked images and masks with the provided script (oxpet_download_and_viz_fixed.py) before training.
Transforms. Images and masks are resized to 384×384 (bilinear for images, NEAREST for masks) and images are normalized with ImageNet statistics. Baseline augmentation is a horizontal flip (p = 0.5). Strong augmentation adds a random resized crop (scale 0.5 to 1.0, aspect 3/4 to 4/3), rotation of ±15° (p = 0.5, rotated-in mask pixels set to background) and color jitter (brightness, contrast and saturation ×0.75 to 1.25, hue ±0.03). Every geometric operation on a mask uses NEAREST interpolation.
Model and training. UNet++ from segmentation_models_pytorch with an ImageNet-pretrained ResNet-34 encoder (26.1M parameters). Loss is pixel-wise cross-entropy (CE), or CE plus multiclass soft Dice with weight 1. AdamW (learning rate 3e-4, weight decay 1e-4), one epoch of linear warmup, then per-step cosine decay to 1% of the peak rate. Batch size 16, 40 epochs, 16-bit mixed precision (AMP). PyTorch Lightning runs the loop; ModelCheckpoint keeps the epoch with the best validation mIoU, and metrics go to CSV and Weights & Biases.
Metrics. A confusion matrix is accumulated over all pixels of a split. Per class, IoU = TP / (TP + FP + FN) and Dice = 2TP / (2TP + FP + FN); mIoU and mDice are the unweighted means over the three classes.
Compute. Each run used one NVIDIA RTX 5070 Ti (16 GB) on the Hamilton CS HPC cluster and took 24 to 25 minutes, 1.25 GPU-hours for all three runs. We used 384×384 rather than 512 because most images are about 500×375, so 384 keeps close to native resolution without upsampling.

Results

Table 1 is the ablation. Each row adds one change to the row above. Validation mIoU is the best epoch; test columns use that checkpoint.

Run	Aug.	Loss	Best val mIoU	Test mIoU	Test mIoU + TTA	Test mDice	Border IoU
Baseline	flip	CE	0.815 (ep. 24)	0.816	0.821	0.891	0.615
+ Strong aug.	strong	CE	0.822 (ep. 37)	0.829	0.833	0.900	0.645
+ Dice loss	strong	CE +	0.821 (ep. 36)	0.829	0.833	0.901	0.647

Run	Aug.	Loss	Best val mIoU	Test mIoU	Test mIoU + TTA	Test mDice	Border IoU
		Dice					

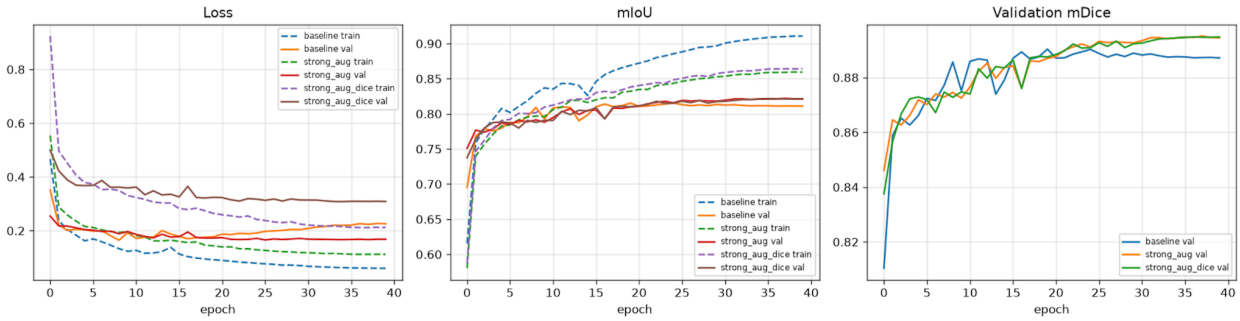
Table 2 gives per-class test scores for the baseline and the final model (strong aug., CE + Dice, flip TTA).

Class	Baseline IoU	Baseline Dice	Final IoU	Final Dice
Pet	0.890	0.942	0.899	0.947
Background	0.942	0.970	0.947	0.973
Border	0.615	0.762	0.654	0.791
Mean	**0.816**	**0.891**	**0.833**	**0.903**

Pixel accuracy is 0.937 for the baseline and 0.942 for the final model.

Learning curves

Figure 1. Loss, mIoU (train dashed, validation solid) and validation mDice per epoch for the three runs.



The baseline overfits. Its validation loss reaches its minimum (0.164) at epoch 8 and rises to 0.226 by epoch 40, and its final train mIoU (0.911) is 0.10 above validation (0.811). With strong augmentation the gap shrinks to 0.04 (0.859 vs 0.821), validation loss stays flat at about 0.17, and the best epoch moves from 24 to 37, so these runs were still improving when training stopped. The CE + Dice validation loss is on a larger scale because it includes the Dice term; it cannot be compared directly with the CE runs.

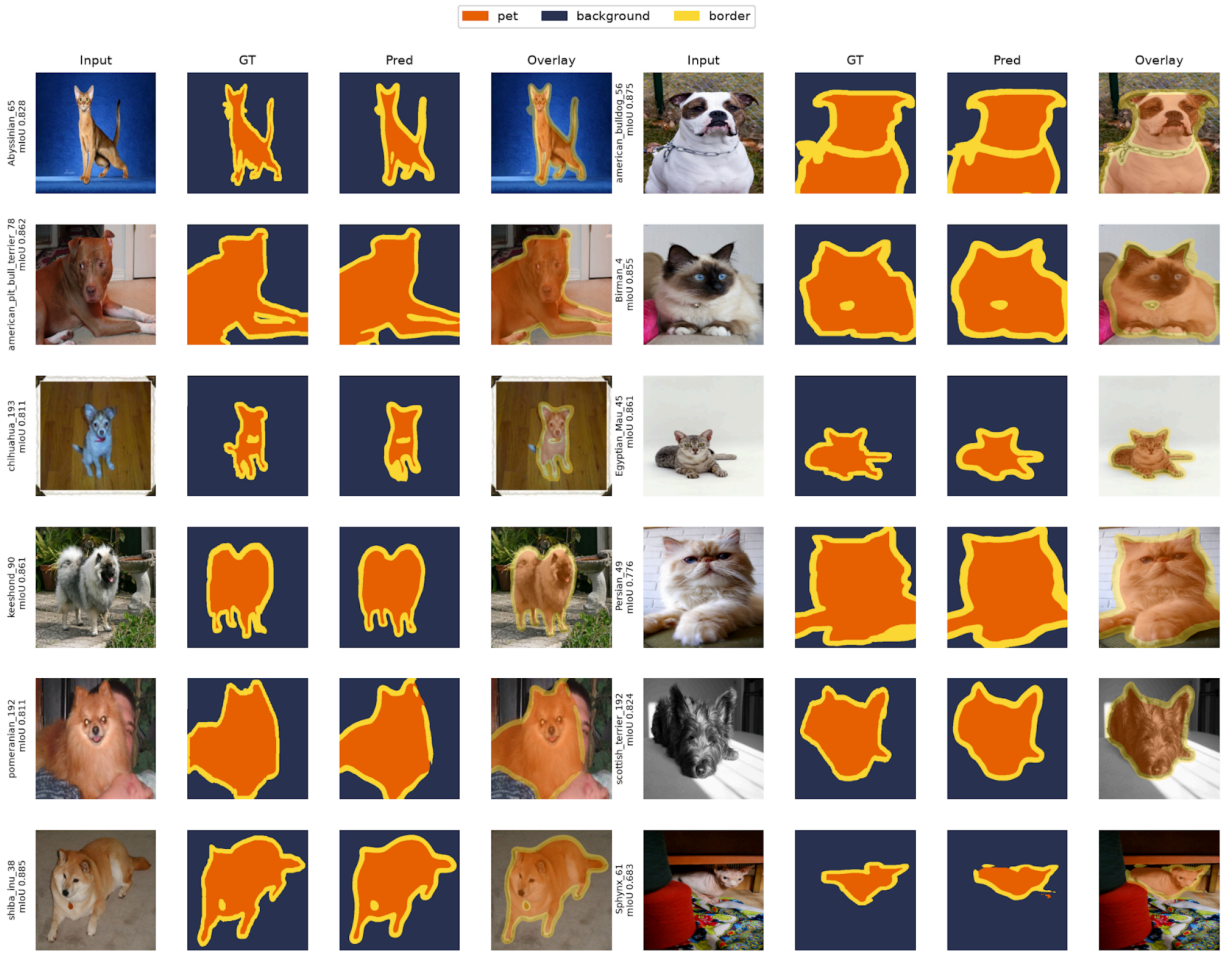
Improvements

Improvement 1: strong augmentation. Random crops, rotations and color jitter raise test mIoU by 1.3 points (0.816 to 0.829) and mDice by 0.9. Most of the gain is on the border class (+3.0 IoU points); pet and background gain 0.5 and 0.3. The curves show why: the baseline memorizes the 2,944 training images after about 15 epochs, while the augmented models keep generalizing to the end of training.
Improvement 2: CE + Dice loss. On top of strong augmentation, adding Dice loss changed test mIoU by +0.05 points and border IoU by +0.3 points, and validation mIoU went slightly down (0.8221 to 0.8214). These differences are smaller than we would expect between two seeds of the same configuration, so we conclude that Dice loss gave no measurable benefit here. Dice loss mainly helps rare classes; in this dataset the smallest class (border) still covers a sizable share of each image, so cross-entropy already trains it well.
Test-time augmentation. Averaging the softmax of the image and its horizontal flip adds 0.4 to 0.6 mIoU points to every model, again mostly on the border class, at twice the inference cost.
Limitations. Each configuration was trained once, so differences below about 0.2 mIoU points (0.002) should not be read as real. The border class remains the weakest (IoU 0.65): it is a thin band whose width

varies between annotations, and the model predicts a band of nearly uniform width.

Qualitative results

Figure 2. Twelve random test images with ground truth, final-model prediction and prediction overlay (orange = pet, dark blue = background, yellow = border). Per-image mIoU is printed next to each input.



The predicted pet regions follow the ground truth closely, including legs and ears, with per-image mIoU from 0.68 to 0.89. The errors fall into three groups. Border width: the model draws a band of even width where the annotators drew a thick band around long fur (Persian_49, 0.78). Thin structures: tails and the gaps between legs are filled or cut off (chihuahua_193, 0.81). Low contrast: the hairless cat in a dark, cluttered scene (Sphynx_61, 0.68) is partly missed and a small false patch appears beside it. The same twelve images for the baseline are in outputs/samples/baseline_pred_grid_1.png and _2.png.

References

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